

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MS-MAE
Program Name	Master of Science in Mechanical and Aerospace Engineering
Assessment Year	2025
Program Assessment Coordinator	Heng Wang
Year of Current Assessment Plan	2025
No Assessment This Year per Plan	☐
LO(s) Assessed	LO1, LO2
Presented at program authority meeting date: <i>Please attach minutes</i>	TBD

Methodology

LO#1

- ⌘ Learning goal assessed: Graduates will be able to apply proper computational and numerical techniques in contexts relevant to Mechanical and Aerospace Engineering.
- ⌘ Sample size: All students completing MMAE 501, MMAE 510
- ⌘ Semesters in which artifacts were collected: Fall, Spring
- ⌘ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions specifically designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.
- ⌘ Proficiency ratings
 - ⌘ scores > 75% points indicate **P** proficient.
 - ⌘ scores > 55% points indicate **M** meets expectations.
 - ⌘ scores < 55% points indicate **D** does not meet expectations.

LO#2

- ⌘ Learning goal assessed: Graduates will demonstrate solid knowledge in a core subject area of their major program.
- ⌘ Sample size: All students completing a subset of MMAE 530, MMAE 541, MMAE 545
- ⌘ Semesters in which artifacts were collected: Fall, Spring
- ⌘ Artifact: For each course, performance of the students in 4 questions chosen from a pool of questions especially designed to assess this learning goal. The questions are delivered to the students in the last third of the semester. Each student will be scored out of 4.

- ∄ Proficiency ratings
 - ∄ scores > 75% points indicate **P** proficient.
 - ∄ scores > 55% points indicate **M** meets expectations.
 - ∄ scores < 55% points indicate **D** does not meet expectations.

LO#3

- ∄ Learning goal assessed: For graduates doing thesis, they will have the ability to conduct research, publish scientific findings, and disseminate that knowledge.
- ∄ Sample size: All students completing their MS thesis.
- ∄ Semesters in which artifacts were collected: Fall, Spring
- ∄ Artifact: A particularly designed questionnaire filled out by the thesis committee after the thesis defense exams. The committee members come to a consensus and rate the students on a scale of 1-5 on each of the following:
 - ∄ Research problem is clearly described
 - ∄ Literature and context of the work are demonstrated
 - ∄ Solution approach, methods and/or tools are described
 - ∄ Results are analyzed, interpreted and communicated well
 - ∄ Written and oral communication are effective
 - ∄ Critical thinking consistent with an independent researcher
- ∄ Proficiency ratings
 - ∄ scores of 4 points and above for all items indicate **P** proficient.
 - ∄ scores of 3 points and above for all items indicate **M** meets expectations.
 - ∄ scores below 3 points on any item indicate **D** does not meet expectations.

Results

Artifact samples

Artifact for LO1 from MMAE 501. Four questions from a bank of 10 were selected and assigned as a quiz. Sample scores listed below. "C" for correct, "I" for incorrect.

1

Multiple Choice 1 point

Given a system of linear differential equations $\frac{dx}{dt} = Ax$, where A is a square matrix, which statement is true about the general solution?

- ☒ If A has repeated eigenvalues, the solution may involve terms of the form $t^k \cdot e^{\lambda t}$
- ☐ It always contains n linearly independent solutions, where n is the size of A
- ☐ It can be expressed as $x(t) = e^{(At)x(0)}$ only if A is invertible
- ☐ The stability of the system depends solely on the determinant of A

MS-MAE LO1 MS-ASAR LO1

3

Multiple Choice 1 point

Consider the Sturm-Liouville problem: $-(p(x)y')' + q(x)y = \lambda r(x)y$ with appropriate boundary conditions. Which statement is correct?

- ☒ The eigenfunctions form an orthogonal set with respect to the weight function $r(x)$
- ☐ The eigenvalues are always real, regardless of the functions $p(x)$, $q(x)$, and $r(x)$
- ☐ The number of nodes in the n th eigenfunction is always exactly n
- ☐ The lowest eigenvalue is always positive if $q(x) > 0$ for all x

MS-MAE LO1 MS-ASAR LO1

2

Multiple Choice 1 point

In the calculus of variations, what is the primary difference between constrained and unconstrained optimization problems?

- ☒ Constrained problems require the use of Lagrange multipliers
- ☐ Constrained problems always lead to simpler Euler-Lagrange equations
- ☐ Unconstrained problems can only be solved using direct methods
- ☐ The extremals for constrained problems are always smooth curves

MS-MAE LO1 MS-ASAR LO1

4

Multiple Choice 1 point

For a complex function $f(z) = u(x,y) + iv(x,y)$, which condition ensures that $f(z)$ is analytic?

- ☒ $u_x = v_y$ and $u_y = -v_x$ (where u_x is the partial derivative of u w.r.t x etc.)
- ☐ $u_x = v_y$ and $u_y = v_x$
- ☐ $u^2 + v^2 = \text{constant}$
- ☐ $u_{xx} + u_{yy} = 0$

MS-MAE LO1 MS-ASAR LO1

	Sample 1	Sample 2	Sample 3
P	C, C, C, C (100%)	C, C, C, C (100%)	C, C, C, C (100%)
M	I, C, C, C (75%)	C, C, C, I (75%)	I, C, C, C (75%)
D	I, C, I, C (50%)	I, C, I, I (25%)	

Artifact for LO2 from MMAE 530. Four questions from a bank of five were selected and assigned as a quiz. Sample scores listed below. "C" for correct, "I" for incorrect.

1
Multiple Choice
1 point

Let α be the coefficient of linear thermal expansion for a linear elastic solid. When this solid is subjected to temperature change, ΔT , which of the following statements is incorrect?

- ☒ Effect of temperature change must be reflected in equilibrium equations
- ☐ Normal strains will increase by an amount equal to $\alpha \Delta T$
- ☐ Temperature change brings a new term to the generalized Hooke's Law
- ☐ Nonuniform temperature change may result in non-zero stresses even when there no mechanical loads applied on the solid

MS-MAE LO2

2
Multiple Choice
1 point

In solving plane elasticity problems using the Airy stress function $\Phi(x,y)$, which of the following is correct?

- ☒ Φ must satisfy $\nabla^2 \Phi = 0$ to ensure equilibrium and compatibility
- ☐ Stress components are obtained from first derivatives of Φ
- ☐ Φ directly gives the displacement field
- ☐ Φ is applicable only for plane stress problems

MS-MAE LO2

3
Multiple Choice
1 point

Consider the deformation gradient tensor F . Which of the following statements is correct?

- ☒ F represents deformation and rigid body rotation
- ☐ $\det(F)$ can be negative under compression
- ☐ F is always symmetric
- ☐ For small deformations, F is approximately equal to the small strain tensor

MS-MAE LO2

4
Multiple Choice
1 point

In the context of stress analysis, which statement about the Cauchy stress tensor σ is true?

- ☒ The hydrostatic stress is given by $(1/3)\text{tr}(\sigma)$, where tr denotes the trace
- ☐ The first invariant of σ is directly related to distortional stresses
- ☐ The eigenvalues of σ represent the normal stresses on the octahedral planes
- ☐ In a state of pure shear, all principal stresses are non-zero

MS-MAE LO2

	Sample 1	Sample 2	Sample 3
P	C, C, C, C (100%)		
M	C, C, C, I (75%)		
D			

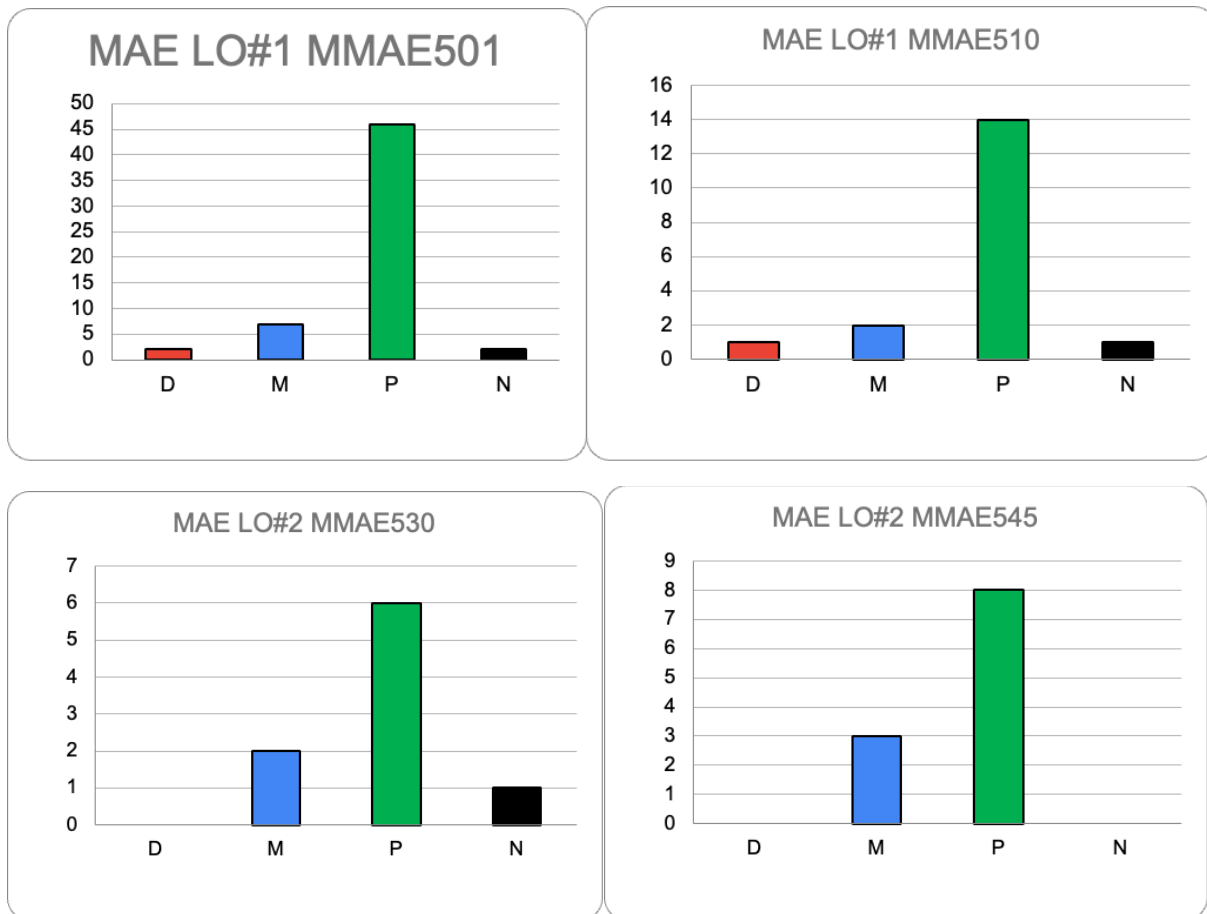
LO3 artifact. Only one MS-MAE student completed thesis defense in F24-S25. This artifact is archived. Analysis will be performed when a sufficient sample has been collected.

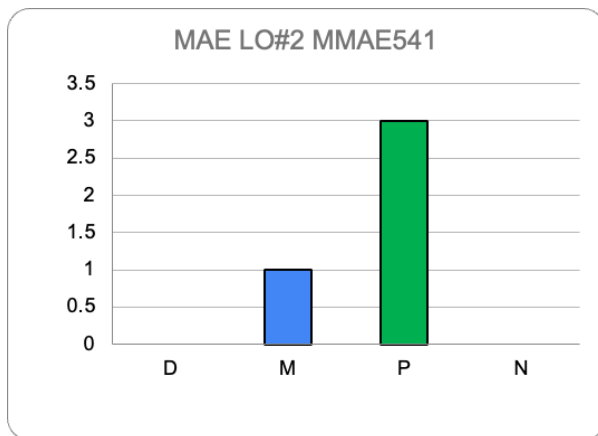
	Sample 1			
Research problem is clearly described	3			
Literature and context of the work are demonstrated	2			
Solution approach, methods and/or tools are described	2			
Results are analyzed, interpreted and communicated well	3			

Written and oral communication are effective	3			
Critical thinking consistent with an independent researcher	3			

Data table

Charts below show the distribution of results from different courses. Y axis is the number of students.





Description of findings

This is the first year with current assessment plan hence no comparison with previous years is available. Nonetheless for both learning objectives the result indicated positive outcomes with most students in the upper category. Especially for LO#1 where a large number of students were assessed. The assessment questions are focused on basic concepts, even for numerical and computational skills. From conventional point of view, this might lack the test on actual problem-solving skills, however, with ever-capable AI tools it can be argued that the essential skill of a person would be mastering the concepts and logics behind methodologies. For LO#2 the number of students assessed is relatively small. Yet across three different courses the same distribution is found: with ~75% students scored perfect scores and ~25% missing one of the four questions which is still acceptable.

There is no concern identified based on current assessment.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, consider the root cause of any deficits, and if areas for possible improvement emerge, describe plans to achieve this improvement.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☒ no previous CI plans available

This cycle's CI plan

Describe any CI plans developed from this cycle's data, with

- Specific curricular or instructional changes in response to this assessment data,
- A timeline for implementation,
- As well a description of who will participate in and manage these plans

☐ After consideration and presentation to the program authority, no CI plans are recommended.

1. Expand current assessment plan to include more courses.
2. While current assessment format with standardized questions is satisfactory, improvements are still preferred including: i) expanding the question banks and assigning more questions. Some of these can probe students' case-specific problem-solving skills, like which method should be used to better solve a problem. ii) collecting additional artifacts from homework or exams that solve specific problems.